Assignment 9

Introduction to Machine Learning

Prof. B. Ravindran

1. Consider the bayesian network shown below.

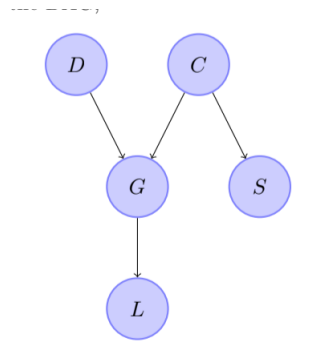


Figure 1: Q1

Two students - Manish and Trisha make the following claims:

- Manish claims $P(D|\{S, L, C\}) = P(D|\{L, C\})$
- Trisha claims $P(D|\{S, L\}) = P(D|L)$

where $P(X|Y)$ denotes probability of event X given Y. Please note that Y can be a set. Which of the following is true?

(a) Manish and Trisha are correct.

(b) Manish is correct and Trisha is incorrect.

(c) Manish is incorrect and Trisha is correct.

(d) Both are incorrect.

(e) Insufficient information to make any conclusion. Probability distributions of each variable should be given.

Sol. (b)
D and S are independent events given two variables $\{L, C\}$ but not in the case when only L is given.

2. Consider the same bayesian network shown in previous question (Figure 1). Two other students in the class - Trina and Manish make the following claims:

- Trina claims $P(S|\{G, C\}) = P(S|C)$
- Manish claims $P(L|\{D, G\}) = P(L|G)$

Which of the following is true?

(a) Both the students are correct.

(b) Trina is incorrect and Manish is correct.

(c) Trina is correct and Manish is incorrect.

(d) Both the students are incorrect.

(e) Insufficient information to make any conclusion. Probability distributions of each variable should be given.

Sol. (a)
same as previous question

3. Consider the Bayesian graph shown below in Figure 2.

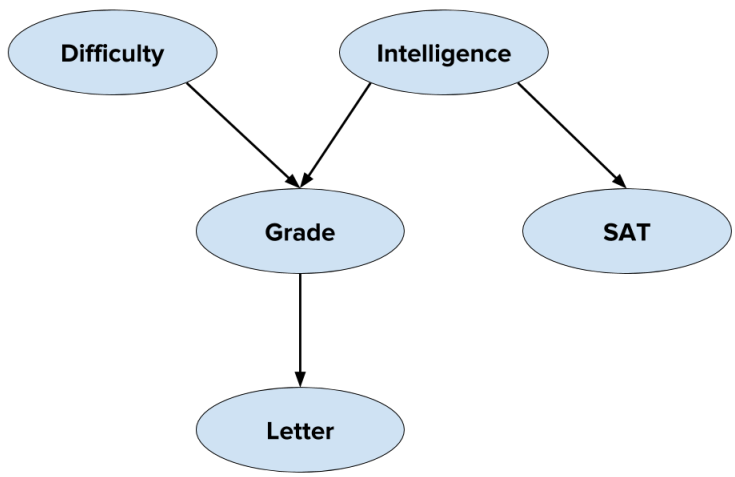


Figure 2: Q3

The random variables have the following notation: d - Difficulty, i - Intelligence, g - Grade, s - SAT, l - Letter. The random variables are modeled as discrete variables and the corresponding CPDs are as below.

d^0	d^1
0.6	0.4

i^0	i^1
0.6	0.4

	g^1	g^2	g^3
i^0, d^0	0.3	0.4	0.3
i^0, d^1	0.05	0.25	0.7
i^1, d^0	0.9	0.08	0.02
i^1, d^1	0.5	0.3	0.2

	s^0	s^1
i^0	0.95	0.05
i^1	0.2	0.8

	l^0	l^1
g^1	0.2	0.8
g^2	0.4	0.6
g^3	0.99	0.01

What is the probability of $P(i = 1, d = 0, g = 2, s = 1, l = 1)$?

(a) 0.004608
(b) 0.006144
(c) 0.001536
(d) 0.003992
(e) 0.009216
(f) 0.007309
(g) None of these

Sol. (e)

$$P(i = 1, d = 0, g = 2, s = 1, l = 0) = P(i = 1)P(d = 0)P(g = 2|i = 1, d = 0)P(s = 1|i = 1)P(l = 1|g = 2)$$
$$= 0.4 * 0.6 * 0.08 * 0.8 * 0.6 = 0.009216$$

4. Using the data given in the previous question, compute the probability of following assignment, $P(i = 1, g = 1, s = 1, l = 0)$ irrespective of the difficulty of the course? (up to 3 decimal places)

(a) 0.160
(b) 0.371
(c) 0.662
(d) 0.047
(e) 0.037
(f) 0.066
(g) 0.189

Sol. (d)

$$P(i = 1, g = 1, s = 1, l = 0) = P(i = 1)P(s = 1|i = 1)P(l = 0|g = 1) \sum_{d}^{d=0,1} (P(d)P(g = 1|i = 1, d))$$

$$P(i = 1, g = 1, s = 1, l = 1) = 0.4 \times 0.8 \times 0.2 \times (0.9 \times 0.6 + 0.5 \times 0.4) = 0.04736$$

5. Consider the Bayesian network shown below in Figure 3

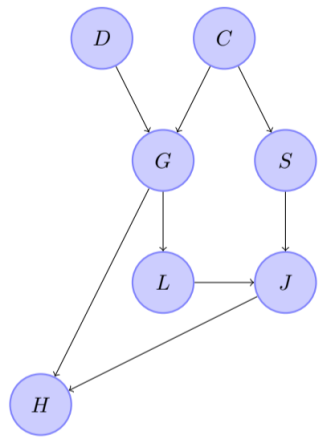


Figure 3: Q5

Two students - Manish and Trisha make the following claims:

- Manish claims $P(H|\{S, G, J\}) = P(H|\{G, J\})$
- Trisha claims $P(H|\{S, C, J\}) = P(H|\{C, J\})$

Which of the following is true?

(a) Manish and Trisha are correct.

(b) Both are incorrect.

(c) Manish is incorrect and Trisha is correct.

(d) Manish is correct and Trisha is incorrect.

(e) Insufficient information to make any conclusion. Probability distributions of each variable should be given.

Sol. (d)

6. Consider the Markov network shown below in Figure 4

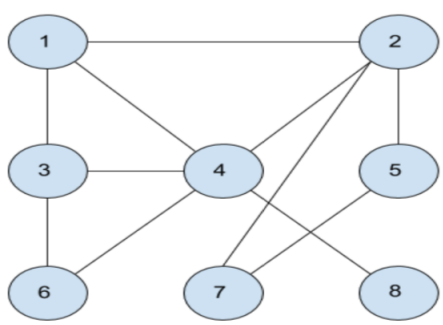


Figure 4: Q6

Which of the following variables are NOT in the markov blanket of variable "3" shown in the above Figure 4 ? (multiple answers may be correct)

(a) 1
(b) 8
(c) 2
(d) 5
(e) 6
(f) 4
(g) 7

Sol. (b), (c), (d) and (g)

7. In the Markov network given in Figure 4, two students make the following claims:

   - Manish claims variable "1" is independent of variable "7" given variable "2".
   - Trina claims variable "2" is independent of variable "6" given variable "3".

   Which of the following is true?

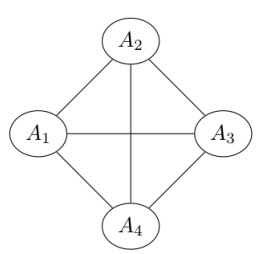
   (a) Both the students are correct.
   (b) Trina is incorrect and Manish is correct.
   (c) Trina is correct and Manish is incorrect.
   (d) Both the students are incorrect.
   (e) Insufficient information to make any conclusion. Probability distributions of each variable should be given.

   Sol. (b)

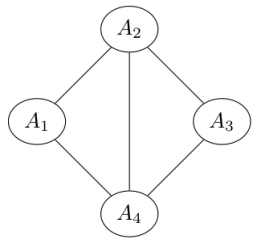
8. Four random variables are known to follow the given factorization

$$P(A_1 = a_1, A_2 = a_2, A_3 = a_3, A_4 = a_4) = \frac{1}{Z}\psi_1(a_2, a_3)\psi_2(a_1, a_4)\psi_3(a_2, a_4)\psi_4(a_1, a_3)$$

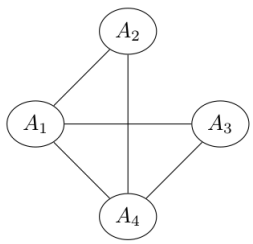
   The corresponding Markov network would be



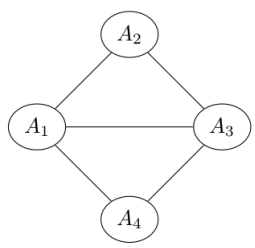
(a)



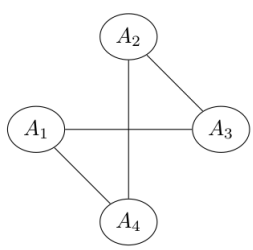
(b)



(c)



(d)



(e)

Sol. (e)

9. Does there exist a more compact factorization involving less number of factors for the distribution given in previous question?

   (a) Yes
   (b) No
   (c) Insufficient information

   Sol. (b)

10. Consider the following Markov Random Field.

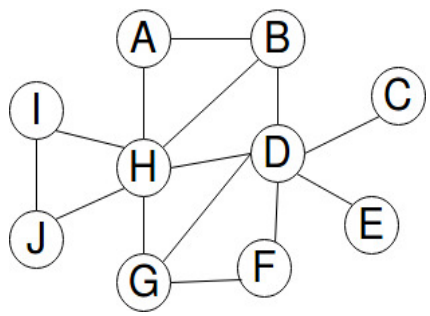


Figure 10: Q10

Which of the following nodes will have no effect on D given the Markov Blanket of D?

   (a) A
   (b) B
   (c) C
   (d) E
   (e) F
   (f) G
   (g) H
   (h) I
   (i) J

   Sol. (a), (h) and (i)
   The question requires you to select the random variables **not** in the Markov blanket of D. We see that the Markov blanket of D contains $B, C, E, F, G, \& H$. The only other variables, other than D are $A, I, \& J$. These three variables can have no effect on D once the Markov blanket is known/given.